ORIGINAL ARTICLE

https://doi.org/10.14730/aaps.2022.00759
Arch Aesthetic Plast Surg 2023;29(2):109-114
pISSN: 2234-0831 eISSN: 2288-9337

aaps
Archives of Aesthetic Plastic Surgery

Achieving attractive and healthy-looking lips through a central lip lift

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Background Lip lifting is a widely accepted and efficient surgical procedure for volumizing the upper lip to achieve a more youthful appearance. A previous study by Hwang et al. identified the lip-related ratios most preferred by Koreans: a lip height-to-width ratio (LTW) of 0.4, an upper-to-lower vermilion ratio (ULR) of 0.8, and an upper vermilion thickness-to-upper lip height ratio (VUL) of 0.5. The current study investigated whether the attractive lip ratio favored by Koreans could be obtained with central upper lip lift surgery.

Methods This retrospective chart review included South Korean patients who underwent central lip lifts between 2015 and 2019 at a single center. The LTW, ULR, VUL, and columellar-labial angle were measured using the patients' preoperative and postoperative clinical photographs.

Results A total of 87 patients underwent central lip lifts. Before surgery, the mean LTW was 0.34, the mean ULR was 0.76, and the mean VUL was 0.30. After surgery, the LTW increased to an average of 0.39, the ULR increased to 0.84, and the VUL increased to 0.393. The patients' postoperative ratio values were close to the attractive lip ratio values proposed by Hwang et al. The average columellar-labial angle decreased from 101.59° to 95.04°, and therefore did not interfere with the aesthetic relationship between the nose and lips.

Conclusions The central lip lift was an efficient cosmetic technique that provided the Korean ideal shape for attractive lips.

Keywords Lip / Lifting / Cosmetic technique

INTRODUCTION

The lips are an important aesthetic landmark on the face. Thick and voluminous lips represent beauty and youth in women, and various cosmetic procedures have been performed to achieve this look. Since the 1980s, surgical methods such as the upper lip lift, Austin-type lip lift, and subnasal lip lift have been introduced [1-3].

In 2019, Hwang et al. [4] determined the length of the upper and lower lips and philtrum that provided the attractive and healthy-looking appearance most preferred by Koreans. They defined attractive and healthy-looking lips using four lip-related ratios, and these results have been applied to aesthetic lip plastic surgery in Koreans.

With the central lip lift, patients can achieve changes in upper vermilion thickness and lip height. Additional cosmetic procedures such as botulinum toxin injection, filler injection, or fat grafting can also improve surgical outcomes [5-7]. In a previous study, Lee et al. [8] confirmed that the height of the upper lip increased significantly as the philtrum length was shortened in central lip lift procedures. Based on the research of Hwang et al. [4] this study investigated whether attractive lip ratios could be obtained in Koreans by performing central lip lifts without additional cosmetic procedures. In addition, the change in the position of the philtrum with the central lip lift affects the columellar-labial angle (CLA). There-

Received: Nov 27, 2022 **Revised:** Jan 9, 2023 **Accepted:** Feb 8, 2023
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fore, we also investigated the aesthetic effects of this unintended CLA change.

METHODS

Study design

A retrospective chart review was conducted on Korean patients who underwent central lip lifts between 2015 and 2019 at a single private clinic. Patients who underwent central lip lifts at the authors' private clinic during this period were included in our study. Patients who had a congenital facial deformity, oral or maxillofacial disease, a history of hypertrophic scars or keloids, excessively downturned lip corners, a history of fillers or botulinum toxin injection, or other cosmetic procedures around the lips and philtrum area were excluded. Patients with a follow-up period of less than 6 months or those who were lost to follow-up were also excluded. Informed consent was obtained from all patients for treatment and for publication of photographs of the postoperative results. This study conformed to the principles of the Declaration of Helsinki. The study was approved by the Institutional Review Board of Yeungnam University Hospital (IRB No. 2021-01-067-002).

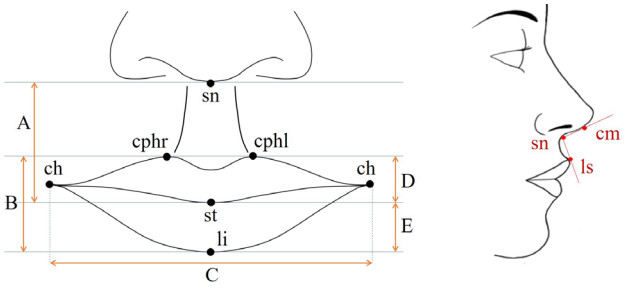


Fig. 1. The lip-related lengths and columellar-labial angle. (A) A, upper lip height, the length between subnasale and stomion; B, lip height, the length between both cristae philtri (right and left) and labium inferius; C, lip width, the length between cheilion; D, upper vermilion thickness, the length between both cristae philtri (right and left) and the stomion; E, lower vermilion thickness, the length between the stomion and labium inferius (sn, subnasale; cphr and cphl, cristae philtri right and left; ch, cheilion; st, stomion; li, labium inferius). (B) Columellar-labial angle, the angle between columella and the line intersecting subnasale and labium superius, cm-sn-ls (cm, columella; sn, subnasale; ls, labium superius).

The patients' frontal and lateral photographs were taken at a consistent distance (1.5 m) using a digital camera (DSLR D7100; Nikon) with the same camera settings and background before surgery and 6 months postoperatively. Adobe Photoshop 7.0 (Adobe Systems, Inc.) was used to measure the number of pixels in the photographs, and the lip-related lengths, ratios, and CLA were calculated (Fig. 1). The lip-related ratios reported by Hwang et al. [4] were calculated using each measured length: lip height-to-width ratio (LTW), upper-to-lower vermilion thickness ratio (ULR), and upper vermilion thickness-to-upper lip height ratio (VUL). The preferred healthy-looking and attractive lip ratios for all ages were reported to be: LTW, 0.4; ULR, 0.8; and VUL, 0.5 (Table 1).

Surgical technique

The philtrum length (the distance from the subnasale to the cristae philtri on both sides) and the lip height were measured using the landmarks described in Fig. 1. Referring to these measured values and in consultation with the patient, we determined how much of the white part of the upper lip was to be resected to obtain the desired lip height. The surgeon then showed the expected results to the patient with a pinch test illustrating the redesigned upper lip and upper teeth in a mirror. In the preoperative design, the upper border of the excision was drawn just inferior to both the columellar base and the nostril sill. Both lateral tips of the excision extended to the uppermost lateral point of the alar-facial groove in the shape of a bull's horn. The lower border of the incision was drawn at intervals below the upper border according to how much of the white part of the upper lip was to be excised. The design of the lower border could be adjusted to maintain a natural contour and symmetry of the upper lip. Based on the preoperative design, en bloc excision of skin, subcutaneous tissue, and partial orbicularis oris was performed under local anesthesia. After surgery, the incisional scar usually remained hidden under the columellar-labial junction and bilateral alar-facial grooves (Fig. 2) [8].

Statistical analysis

For anthropometric assessment of the 87 patients who underwent a central lip lift, the preoperative and postoperative lip-related ratios and CLA measurements were assessed statistically using the paired t-test with SPSS software version 13.0 (SPSS, Inc.). Statistical significance was set at $P < 0.05$.

Table 1. The definition and ideal values of the lip-related ratios preferred by Koreans

Lip-related ratio	Definition	The healthy-looking and attractive lip ratio[a]
Lip height-to-width ratio	Lip height (B)/lip width (D)	0.4
Upper-to-lower vermilion ratio	Upper vermilion thickness (D)/lower vermilion thickness (E)	0.8
Upper vermilion thickness-to-upper lip height ratio	Upper vermilion thickness (D)/upper lip height (A)	0.5

A: upper lip height, B: lip height, C: lip width, D: upper vermilion thickness, and E: lower vermilion thickness in Fig. 1.
[a]Ratio of according to Hwang et al. [4].

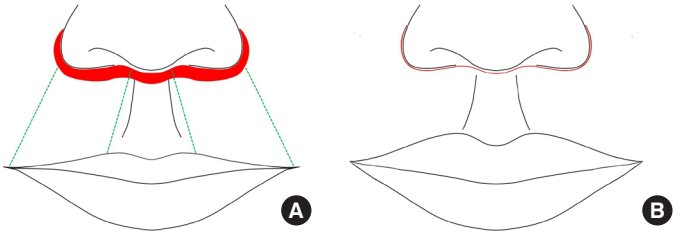


Fig. 2. Schematic illustration demonstrating the preoperative design of a central lip lift. (A) Preoperative illustration of the central lip lift. Both lateral tips of the excision were extended to the uppermost lateral point of the alar-facial groove in the shape of a bull's horn. The lower border of the incision was drawn at intervals below the upper border according to the amount needed to excise the white part of the upper lip. According to preoperative design, en-bloc excision of the skin, subcutaneous tissue, and partial orbicularis oris was performed (red zone). (B) Postoperative illustration of the central lip lift with shortened philtrum height and increased height of the upper lip.

RESULTS

Of 102 patients, 87 satisfied the inclusion criteria. There were 10 men and 77 women with a mean age of 35.6 years (range, 19–67 years). The mean follow-up was 24.2 months (range, 6.0–31.2 months). The preoperative and postoperative patient distributions are presented in Table 2. The preoperative mean was 0.34 for LTW, 0.76 for ULR, 0.30 for VUL, and 101.59° for CLA. After surgery, the mean respectively changed to 0.39 for LTW, 0.84 for ULR, 0.39 for VUL, and 95.04° for CLA. The changes in all lip-related ratios before and after surgery were statistically significant when compared using the paired t-test (Table 3). Furthermore, all lip-related ratios were close to the attractive values identified by Hwang et al. [4].

DISCUSSION

"Pouting" lips are considered symbols of beauty and youth. The lips undergo morphological and histological changes with age, including elongation of the upper lip height, a decrease in the visible vermilion, thinning of the epidermis and dermis, and atrophy of the orbicularis oris muscle. These changes result in a total volume loss of the lips [9,10].

In recent years, various upper lip augmentation and lip lift techniques have been introduced to overcome the effects of aging and make the lips more attractive. Permanent injectable fillers and lip implants can achieve great effects through simple procedures, but there are possible side effects, such as infection, foreign body granuloma, and allergic reaction [11,12]. Free fat grafting has been a widely used and effective method without such side effects, but further studies are needed to assess whether it can be maintained long-term [7,13,14]. A mucosal advancement technique increases the vermilion height with the scar hidden but has the potential for numbness around the lips [15]. Since Rozner and Isaacs [16] first performed the lip lift in 1981, several techniques such as the Austin-type, scarless upper lip suspension, and subnasal lip lift have been developed [1-3,17]. The authors' central lip lift successfully increased the height of the visible upper vermilion while reducing the philtrum length (Fig. 3) [8]. Austin and Weston [2] reported that raising the central part of the upper lip may cause the patient to look sad by making the edge of the lips turn down. This phe-

Table 2. Preoperative and postoperative distribution of patients according to the range of lip-related ratios

Lip-related ratio	Range	Patients (n=87)	
		Preoperative	Postoperative
LTW	<0.25	5	0
	0.25–0.33	37	19
	0.33–0.40	33	32
	0.40–0.50	11	28
	>0.50	1	8
ULR	<0.60	11	6
	0.60–0.75	29	17
	0.75–0.80	16	10
	0.80–1.00	26	44
	>1.00	5	10
VUL	<0.33	61	13
	0.33–0.40	24	35
	0.40–0.50	2	36
	0.50–0.80	0	3
	<0.80	0	0
CLA (°)	<80	0	6
	80–90	12	18
	90–100	20	31
	100–110	46	27
	>110	9	5

LTW, lip thickness-to-width ratio; ULR, upper-to-lower vermilion ratio; VUL, upper vermilion thickness-to-upper lip height ratio; CLA, columellar-labial angle.

Table 3. Comparison of lip-related ratios before and after central lip lift

Lip-related ratio	Preoperative average	Postoperative average	P-value
LTW	0.34	0.39	<0.00
ULR	0.76	0.84	<0.00
VUL	0.30	0.39	<0.00
CLA (°)	101.59	95.04	<0.00

LTW, lip thickness-to-width ratio; ULR, upper-to-lower vermilion ratio; VUL, upper vermilion thickness-to-upper lip height ratio; CLA, columellar-labial angle.

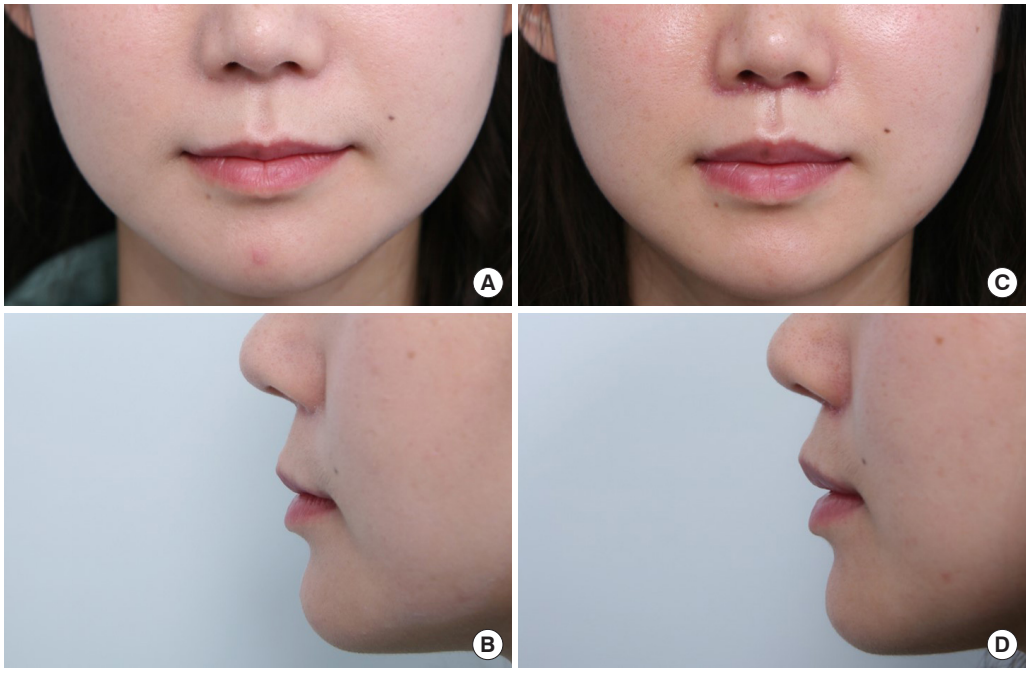


Fig. 3. Preoperative (A, B) and postoperative (C, D) clinical photographs of a 35-year-old female patient who underwent a central lip lift. The postoperative photographs were taken 6 months after surgery. The lip thickness-to-width ratio, upper-to-lower vermilion ratio, and upper vermilion thickness-to-upper lip height ratio were measured in a frontal view photograph. The columellar-labial angle was measured in a lateral view photograph. (C, D) The height of the visible upper vermilion was increased while reducing the philtrum length with a central lip lift. The incisional scar usually remained hidden under the columellar-labial junction and bilateral alar-facial grooves.

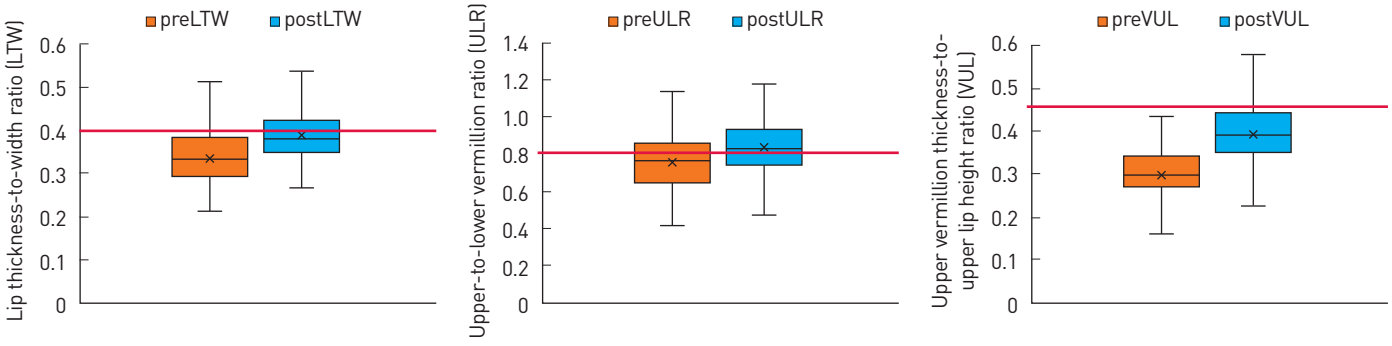


Fig. 4. Change of lip-related ratios. The mean value of the LTW increased from 0.335 to 0.389 postoperatively. The mean values of the ULR and the VUL increased from 0.757 to 0.837 and from 0.299 to 0.393, respectively. All mean values of the LTW, ULR, and VUL approached the attractive values (red line, 0.4 for LTW, 0.8 for ULR, and 0.5 for VUL).

nomenon is called the "sad appearance deformity." To prevent this, we extended the incision line to the lateral aspect of the alar so that the lateral aspect of the lips was slightly elevated. The upper vermilion was everted upward by removing the skin and muscle right below the columellar-labial junction in the central lip lift. As the lip everts, the incisor edges of the upper teeth may be exposed.

Several studies have been published on ideal lip proportions. Penna et al. [9] showed that the attractive ratio of the height of the upper vermilion to the length between the lips and nose was 0.28 for males and 0.28 for females. However, Asians tend to have a smaller oral width and retruded chin than Caucasians and the lips of Asians are more prominent [18]. Therefore, healthy-looking and attractive lips may differ according to race. Hwang et al. [4] surveyed 258 Korean women and 72 Korean men to rate four lip ratios for attractive, healthy-looking lips. The preferred lip ratios were LTW, 0.4; ULR, 0.8; and VUL, 0.5. The current study cited Hwang's research to determine whether the results after central lip lift corresponded to the ideal lip shape from a Korean perspective. After the

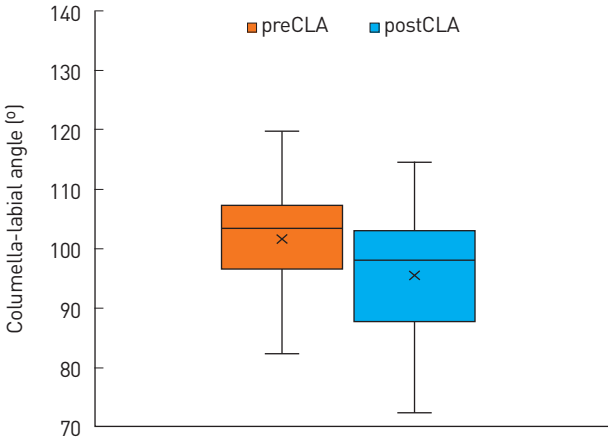


Fig. 5. Change of the columellar-labial angle. The average columellar-labial angle (CLA) decreased from 101.59° to 95.04°. The CLA change after performing the central lip lift was within the ideal ranges mentioned in several studies.

central lip lift, the LTW, ULR, and VUL significantly increased to 0.39, 0.84, and 0.94, respectively, which were close to the attractive values identified by Hwang et al. (Fig. 4) [4].

The overall average CLA decreased from 101.59° preoperatively to 95.04° postoperatively. The average CLA for the 10 men before surgery was 100.22°, while for the 77 women it was 101.79°. Both average values significantly decreased postoperatively to 91.41° and 96.05°, respectively. In other studies, the ideal CLA value varied, and different patterns for the range of the preferred angle were shown. Powell reported that the ideal CLA was 90° to 120°, and Maran and Lund reported 90° to 105° [19,20]. Armijo et al. [21] reported that the ideal CLA was 93.4° to 98.5° in men and 95.5° to 100.1° in women. The CLA change in our study after the central lip lift was within the ideal ranges (Fig. 5). The central lip lift did not disturb the aesthetic relationship between the nose and lips.

In the authors' previous study analyzing the results of central lip lifts, complications were relatively minor, with 0.99% of patients requiring revisional treatment due to wound dehiscence and hematoma, and 2.5% requiring revisional surgery due to an asymmetric upper lip [8]. In the current study, no significant complications required revisional treatment. Twelve patients showed minimal wound dehiscence that recovered fully with conservative treatment. Twenty-five patients showed partial exposure of the upper teeth as their upper lip was lifted, which may affect the aesthetic satisfaction of the patients after surgery. The postoperative scars required careful management for aesthetic reasons. Fractional CO_2 laser treatments were performed on the visible postoperative scars. Although this study excluded patients who had received intraoperative botulinum toxin injection into the muscle, its effect on scarring reduction has been demonstrated in several studies [5,6,22]. In patients not included in this study, botulinum toxin was injected after central lip lifts to prevent scar enlargement by reducing muscular distortion and minimizing skin tension.

Our study had several limitations. First, there was a difference in the numbers of male and female patients, with only 10 male patients included in the study. Second, the lip-related ratios and CLA were calculated from clinical photographs using Adobe Photoshop 7.0, not from actual patients. Therefore, the measured values might contain errors relative to the actual values. Finally, few previous studies, apart from the research of Hwang et al. [4], could be used as standards for the ideal lip ratio of Koreans.

In this study, we attempted to quantitatively and objectively evaluate the effect of the central lip lift based on the attractive and healthy-looking lip ratio proposed by Hwang et al. [4] The upper vermilion height increased and the philtrum height decreased, approaching the ideal aesthetic ratio with the central lip lift. The CLA did not interfere with the aesthetic relationship between the nose and lips. In conclusion, the central lip lift can be an efficient cosmetic technique to create attractive lips for Koreans.

NOTES

Conflict of interest

No potential conflict of interest relevant to this article was reported.

Ethical approval

The study was conducted in accordance with the Declaration of Helsinki. The study was approved by the Institutional Review Board of Yeungnam University Hospital (IRB No. 2021-01-067-002).

Patient consent

The patients provided written informed consent for the publication and use of their images.

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